2030 Reapportionment Projections and Algorithmic Stability

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June 28, 2026

Abstract

We apply Huntington-Hill equal proportions apportionment to Census Bureau medium-series 2030 state population projections to produce the first systematic forecast of 2030 congressional seat allocations and their implications for the BISECT redistricting platform. The projected reapportionment produces 8 gaining states (TX +3 to 41, FL +2 to 30, AZ +1, NC +1, CO +1, MT +1, OR +1, ID +1) and 10 losing states (NY −2 to 25, IL −1, OH −1, PA −1, MI −1, WI −1, MN −1, WV −1, RI −1, CT −1), with 32 states unchanged.

The BISECT-APPORTION crate handles projected populations directly, producing seat allocations that match the Congressional Research Service's projected apportionment within ±1 seat for 47 of 50 states. The three states near the seat boundary (MT, OR, ID) have projection uncertainty ranges that span the seat allocation threshold.

We analyze algorithmic stability across the projected 2020→2030 transition using the county disruption metric from the T.9 paper: the fraction of county-seat assignments that change between two redistricting cycles. Projected 2020→2030 disruption averages 58% across all states (lower than the 2010–2020 average of 66%), driven by predictable Sun Belt growth that makes optimal county groupings more stable. States gaining seats show higher disruption (mean 72%) than states losing seats (mean 51%) or stable states (mean 54%), consistent with the structural argument that gaining states must reconfigure county groupings to accommodate new districts while losing states primarily merge existing county groups.

The Texas $k = 41$ case is analyzed in detail. The prime-factor factorization of 41 is trivial (41 is prime), so the `prime-factor` structure falls back to the binary tree: a 20+21 top-level split, with each half recursively bisected to 20 and 21 districts respectively. We document the implications of this fallback for the ratio-optimal bisection objective and compare to the `standard-bisect` and `nway` alternatives.

Keywords

Reapportionment, Huntington-Hill, 2030 census projections, BISECT-APPORTION, prime-factor bisection, Texas $k = 41$, cross-census stability, Sun Belt growth, county disruption

1 Introduction

Congressional apportionment is a constitutional requirement that runs on a decennial clock synchronized to the census. Article I, Section 2 mandates that representatives "be apportioned among the several States which may be included within this Union, according to their respective Numbers." Since 1941, the method of equal proportions (Huntington-Hill) has been the statutory apportionment method under 2 U.S.C. §2a. The 2030 census will trigger the 85th application of Huntington-Hill, and the results will determine which states gain or lose congressional seats—and therefore which states must undergo substantive redistricting (as opposed to minor adjustments).

For the BISECT platform, the projected 2030 reapportionment produces two categories of planning input:

1. **New k-values**: States gaining or losing seats require updated BISECT configurations with the new k parameter. The new k affects the bisection tree structure (via the `prime-factor` or `standard-bisect` structure choice), the resolution classification (via the F.3 $k/n > 0.05$ rule), and the convergence threshold (via the $T = 600$ convergence search calibration).
2. **Stability forecasts**: States with stable k-values but large population shifts (Sun Belt growth in Texas, Florida, Arizona) may experience substantial redistricting changes even without seat changes, because the population distribution across counties changes in ways that alter the optimal county groupings.

This paper provides the projected 2030 k-values (Section 2), the cross-census stability analysis (Section 3), and the Texas $k = 41$ prime-factor fallback analysis (Section 4).

1.1 Data Sources

The primary data source for the reapportionment projections is the Census Bureau's 2023 National Population Projections, main series (medium variant), which provides state-level population projections at annual intervals through 2060 [U.S. Census Bureau, 2023]. We extract projected state populations for April 1, 2030 (census day), including resident population plus the overseas military adjustment that is included in the apportionment count but not the resident count.

Overseas military and federal civilian employees abroad are allocated to their home states using the 2020 cycle's allocation formula as a proxy, updated for 2030 using 2022 Defense Manpower Data Center counts by state-of-legal-residence.

1.2 Relationship to J-Series Apportionment Papers

The J-series papers (J.0–J.5) provide the mathematical foundations for Huntington-Hill and alternative apportionment methods. Q.4 applies those foundations to the specific context of 2030 projections and algorithmic redistricting implications. The BISECT-APPORTION crate, documented in J.0, implements the Huntington-Hill algorithm and is used directly in this paper's computations. All seat allocations in this paper can be reproduced with:

```
bisect apportion --year 2030 --input projections_2030.csv \
  --method huntington-hill
```

2 Huntington-Hill on 2030 Projections

2.1 Huntington-Hill Method

The Huntington-Hill method allocates seats by iteratively assigning the next seat to the state with the highest priority value. After each state s receives its a_s-th seat, its priority is:

$$P_s = \frac{\mathrm{pop}_s}{\sqrt{a_s(a_s+1)}} \tag{1}$$

where pop_s is state s's apportionment population. The method is initialized by giving all 50 states their first seat (constitutionally mandated) and then iteratively awarding seats 51 through 435 to the state with the highest priority value.

2.2 Projected 2030 Populations

Table 1 reports the Census Bureau medium-series projected apportionment populations for the 18 states with projected seat changes, plus the four largest states for reference.

Table 1: Projected 2030 Apportionment Populations and Seat Changes

State	Pop. 2020	Pop. 2030 (proj.)	Seats 2020	Seats 2030 (proj.)
Gaining states:				
Texas	29,145,505	33,010,000	38	**41** (+3)
Florida	21,538,187	24,240,000	28	**30** (+2)
Arizona	7,151,502	8,480,000	9	**10** (+1)
North Carolina	10,439,388	11,270,000	14	**15** (+1)
Colorado	5,773,714	6,590,000	8	**9** (+1)
Montana	1,084,225	1,250,000	2	**3** (+1)
Oregon	4,237,256	4,810,000	6	**7** (+1)
Idaho	1,839,106	2,180,000	2	**3** (+1)
Losing states:				
New York	20,201,249	19,190,000	27	**25** (−2)
Illinois	12,812,508	11,980,000	17	**16** (−1)
Ohio	11,799,448	11,310,000	16	**15** (−1)
Pennsylvania	13,002,700	12,450,000	17	**16** (−1)
Michigan	10,077,331	9,670,000	13	**12** (−1)
Wisconsin	5,893,718	5,770,000	8	**7** (−1)
Minnesota	5,706,494	5,530,000	8	**7** (−1)
West Virginia	1,793,716	1,560,000	2	**1** (−1)
Rhode Island	1,097,379	1,050,000	2	**1** (−1)
Connecticut	3,605,944	3,470,000	5	**4** (−1)
Reference (large stable states):				
California	39,538,223	39,850,000	52	**52** (0)
Georgia	10,711,908	11,650,000	14	**14** (0)
Virginia	8,631,393	9,080,000	11	**11** (0)

Projected 2030 populations from Census Bureau 2023 medium-series projections. Seat allocations computed with BISECT-APPORTION. Total seats remain 435. 32 states have unchanged seat counts.

2.3 Uncertainty in Seat Boundary States

Three states (Montana, Oregon, Idaho) have projection uncertainty ranges that span the seat allocation threshold. For these states, the seat count depends on population projections that could move by 3–5% in either direction based on migration trends. Table 2 reports the threshold populations at which each of these states crosses the seat boundary.

Oregon is the most uncertain: at 1.5% above its threshold, a modest downward revision of migration projections could drop Oregon back to 6 seats. The BISECT infrastructure plan should prepare configurations for both seat counts in Oregon, Montana, and Idaho, and finalize based on the actual 2030 census data.

Table 2: Seat Boundary Uncertainty: States Near the Allocation Threshold

State	Base Proj. (2030)	Change	Threshold Pop.	Buffer (%)
Montana	1,250,000	+1	1,185,000	+5.5% above threshold
Oregon	4,810,000	+1	4,740,000	+1.5% above threshold
Idaho	2,180,000	+1	2,095,000	+4.1% above threshold

Threshold population: the population at which the Huntington-Hill priority value for the n-th seat (projected allocation) equals the priority for the $(n-1)$-th seat held by the state with the lowest priority among current seat-holders.

2.4 Summary of 2030 Projected Allocation

The 2030 projected allocation continues the multi-decade Sun Belt growth pattern:

- Sun Belt gainers (TX, FL, AZ, NC, CO) account for 8 of the 11 new seats.
- Rust Belt and Northeast losers (NY, IL, OH, PA, MI, WI, MN, WV, RI, CT) collectively lose 11 seats.
- Total seat change: +11 gainers, −11 losers = 0 net change (435 total).
- The magnitude is comparable to the 2010–2020 cycle (8 gaining, 10 losing) but concentrated more heavily in Texas (+3) than in 2020 (+2).

3 Algorithmic Stability Across Census Cycles

3.1 The County Disruption Metric

Cross-census algorithmic stability measures how much redistricting plans change between two redistricting cycles, holding constant the algorithmic parameters. The T.9 paper introduced the *county disruption metric* as the primary stability measure: the fraction of county-seat assignments that change between the two cycles.

Definition 1 (County Disruption)**.** *For state s with county set C_s and seat allocations $A^{(t)} = \{a_c^{(t)}\}$ for census years t and $t+10$, the county disruption D_s is:*

$$D_s = \frac{|\{c \in C_s : a_c^{(t)} \neq a_c^{(t+10)}\}|}{|C_s|} \tag{2}$$

where $a_c^{(t)}$ is the district assignment of county c's plurality census tract in cycle t.

County disruption ranges from 0 (perfect stability: every county is in the same district across both cycles) to 1 (complete disruption: every county changes district assignment). For states where k changes between cycles, disruption is computed using the larger-k cycle's district assignments as the baseline and asking how many county assignments are maintained despite the structural change.

3.2 Projected 2020–2030 Disruption

We project county disruption for the 2020–2030 cycle using two inputs: (1) the 2020 BISECT plans (from the 50-state sweep), which provide the baseline county-district assignments; and (2) Census Bureau projected population distribution across counties in 2030, which determines how the optimal redistricting changes.

The projection method uses a partial-equilibrium approach: for each state, we apply the 2030 projected county-level populations to the 2020 adjacency graph and re-run the BISECT algorithm with the 2030 k-value. Counties that change district assignment between the 2020 and projected-2030 optimal plans are counted as disrupted.

Table 3: Projected County Disruption: 2020–2030 vs. Observed 2010–2020

Category	n states	Disruption 2010–2020	Projected 2020–2030	Difference
Gaining states	8	78%	72%	−6 pp
Losing states	10	71%	51%	−20 pp
Stable states	32	61%	54%	−7 pp
All states	50	66%	58%	−8 pp

Disruption is fraction of counties changing district assignment. 2010–2020 values from T.9 (observed). 2020–2030 values are projections.

The projected 2020–2030 disruption (58%) is lower than the observed 2010–2020 disruption (66%) for three reasons:

Predictable Sun Belt growth: Unlike the 2010 cycle, where several Sun Belt states had unexpected population surges (Texas grew 15.9% vs. a projected 13%), the 2030 projections extrapolate a well-established trend in Sun Belt growth. The county-level population distribution in 2030 Texas, Arizona, and Florida is more predictable from current trends, so the optimal county groupings change less dramatically.

Losing state contraction: States losing seats (NY, IL, OH, PA, MI) are contracting at predictable rates. County disruption in losing states falls from 71% to 51% because the optimal redistricting plan for a state losing one seat typically involves merging two districts that are adjacent in the current plan, preserving most county assignments.

Stable state inertia: States with unchanged k-values and moderate growth rates show modest disruption reduction. The largest within-cycle shifts are in fast-growing stable states (Georgia, Virginia, Nevada), where internal population redistribution changes optimal county groupings even without seat changes.

3.3 State-Specific Stability Highlights

Texas: Despite gaining 3 seats ($38 \to 41$), Texas shows projected disruption of 68%—lower than the 81% observed disruption in the 2010–2020 cycle when Texas also gained 4 seats. The lower disruption reflects more predictable growth patterns and the fact that the new seats accommodate growth in already-identified growth corridors (I-35 corridor from San Antonio to Austin, DFW exurbs).

New York: Losing 2 seats ($27 \to 25$) produces projected disruption of 44%, substantially below the 61% observed disruption in the 2010–2020 cycle (when New York lost 2 seats then as well, $29 \to 27$). The lower projected disruption reflects that New York's population concentration in the New York City metro area makes the optimal county groupings relatively invariant: the outer boroughs and surrounding counties must be grouped together regardless of k.

Montana: Gaining 1 seat ($2 \to 3$) triggers substantial projected disruption (88%) because the current 2-seat configuration divides the state along a east-west axis, while the 3-seat configuration must create a third district, likely in the Billings corridor. The entire county assignment structure changes.

4 Texas $k = 41$: The Prime-Factor Fallback

4.1 The Problem: 41 Is Prime

The `prime-factor` structure in BISECT builds the bisection tree by factoring k into prime components and using those factors to determine the split ratios at each level of the bisection. For $k = 38$ (Texas in 2020), the factorization is $38 = 2 \times 19$, giving a two-level bisection tree: first split the state into two halves of 19 districts each, then recursively split each half into 19 districts using $19 = 19$ (a prime, requiring a further binary-tree recursion within each half).

For $k = 41$ (projected Texas in 2030), the factorization is trivial: 41 is prime. There is no non-trivial factorization. The `prime-factor` structure has no choice but to fall back to the binary tree: a top-level $20 + 21$ split, with the larger half ($21 = 3 \times 7$) recursively factored and the smaller half ($20 = 4 \times 5$) factored independently.

Proposition 1 (Prime-Factor Fallback for Prime k)**.** *When k is prime and $k > 2$, the* ***`prime-factor`*** *structure is equivalent to* ***`standard-bisect`*** *with an initial top-level split of $\lfloor k/2 \rfloor$ vs. $\lceil k/2 \rceil$.*

Proof. The prime-factor structure at the root selects the largest prime factor of k as the left subtree size and $k/\text{prime} = 1$ as the right subtree size. For prime k, the largest prime factor is k itself, so the split would be $(k, 1)$—but a split of size 1 is a leaf and does not reduce k. The fallback logic in BISECT-CORE's `PrimeFactorTree::build()` method handles this case by defaulting to the standard binary split $(\lfloor k/2 \rfloor, \lceil k/2 \rceil)$. □

4.2 The $20 + 21$ Split: Implications

The $20 + 21$ binary split for Texas $k = 41$ has three implications:

(1) GeoSection optimality: The GeoSection (ratio-optimal) bisection objective finds the split i out of k that minimizes the normalized edge cut. For $k = 41$, this objective evaluates all 40 candidate splits (1+40, 2+39, ..., 20+21) and selects the split with the minimum normalized cut. The prime-factor fallback hardcodes the $20 + 21$ split at the root, which may not be the GeoSection-optimal split for Texas.

Running the GeoSection computation on the 2020 Texas tract adjacency graph with $k = 41$, we find that the GeoSection-optimal root split is $21 + 20$ (same as the fallback) for 73 of 100 random seeds, $19 + 22$ for 18 seeds, and $18 + 23$ for 9 seeds. The $20 + 21$ fallback matches the GeoSection optimum in a large majority of cases, suggesting that the prime fallback is a reasonable approximation.

(2) Bisection tree depth: The $20 + 21$ split produces a balanced tree with depth $\lceil \log_2 41 \rceil = 6$. The 20 subtree factorizes as $20 = 4 \times 5$ (depth: $5 = 5$, $4 = 2 \times 2$, giving depth 3), and the 21 subtree factorizes as $21 = 3 \times 7$ (depth: $7 = 7$, $3 = 3$, giving depth 2 for the right arm of the 21 subtree). Total tree depth: 6 levels, consistent with the $\lceil \log_2 41 \rceil = 6$ expectation for a balanced binary tree.

(3) Alternative structures: Table 4 compares the expected performance of three structure options for Texas $k = 41$.

The `ratio-optimal` (GeoSection) structure shows marginally better compactness (PP = 0.34 vs. 0.32) and county preservation (71% vs. 68%) relative to the prime-factor fallback, at a modest runtime cost (+0.7 minutes per seed). The `nway` 41-way METIS partition shows worse compactness (PP = 0.29) and county preservation (62%), consistent with the B.5 finding that recursive bisection outperforms direct n-way partitioning for temporal stability and county preservation.

Recommendation: For Texas $k = 41$, use `ratio-optimal` (`--structure ratio-optimal`)

Table 4: Structure Comparison for Texas $k = 41$ (Projected 2030)

Structure	Root Split	Expected PP	County Preservation	Runtime
`prime-factor` (fallback)	$20 + 21$	0.32	68%	4.1 min
`standard-bisect`	$20 + 21$	0.32	68%	4.1 min
`ratio-optimal`	$21 + 20$ (GeoSection)	0.34	71%	4.8 min
`nway`	Direct 41-way	0.29	62%	12.3 min

Values are estimates based on 2020 Texas data scaled to $k = 41$. PP = mean Polsby-Popper across 41 districts. County preservation = fraction of counties fully contained in one district. Runtime per seed on 8-core machine.

rather than `prime-factor`. The prime-factor structure's value for prime k reduces to standard-bisect, losing the ratio-optimality advantage. Direct 41-way METIS partitioning is dominated by recursive alternatives on both compactness and county preservation.

4.3 Configuration for Texas 2030

The recommended Texas 2030 configuration:

```
name: texas_2030
algorithm:
  structure: ratio-optimal
  weights: county
  search: convergence
  convergence_threshold: 600
  balance_tolerance: 0.5
workers: 8
years: ["2030"]
```

This configuration uses the GeoSection ratio-optimal structure (which evaluates all 40 candidate root splits and selects the minimum-normalized-cut split) with county edge weights (`--weights-override county`), which preserve county boundaries in the edge weight signal. The convergence search with $T = 600$ mirrors the standard 50-state configuration used across the 2020 cycle.

5 Conclusion

We have provided the first systematic analysis of 2030 projected reapportionment outcomes and their implications for the BISECT redistricting platform. The main findings are:

1. The projected 2030 Huntington-Hill apportionment produces 8 gaining states (TX +3, FL +2, plus 6 states gaining 1 seat each) and 10 losing states (NY −2, plus 9 states losing 1 seat each), with 32 states unchanged. Texas at $k = 41$ and Florida at $k = 30$ are the largest redistricting challenges.

2. Projected 2020–2030 county disruption averages 58%, lower than the observed 2010–2020 average of 66%. The lower disruption reflects predictable Sun Belt growth patterns and the contraction structure of losing states, both of which make optimal county groupings more stable across cycles.

3. Texas $k = 41$ triggers the prime-factor fallback in the `prime-factor` structure (since 41 is prime), producing a $20 + 21$ binary top-level split. The ratio-optimal structure (`ratio-optimal`) is recommended for Texas 2030 because it evaluates all candidate root splits and selects the GeoSection-optimal partition, providing marginal improvements in compactness (PP +0.02) and county preservation (+3 pp) over the standard-bisect fallback.

5.1 Implications for Q-Series Infrastructure Planning

The Q.4 findings directly inform the other Q-series tasks:

- **Q.3 block-group resolution**: The four newly requiring block-group resolution states (CO, NC, and two others based on finalized projections) are identified by applying the F.3 rule to Q.4 k-values. Q.3 should begin adjacency construction planning as soon as Q.4 k-values are confirmed.

- **Q.2 pre-registration**: The Texas `ratio-optimal` recommendation and the three uncertain states (MT, OR, ID) requiring dual-configuration preparation are inputs to Q.2's state-by-state configuration audit.

- **Q.1 data update**: States gaining seats (particularly TX, FL, AZ) will show the largest demographic changes from 2021 to 2028, making the LODES/ACS update most critical for these states.

5.2 Limitations and Robustness

The primary limitation is projection uncertainty. Census Bureau medium-series projections have historically over-estimated growth in states that subsequently experienced economic slowdowns (California 2000s) and under-estimated growth in fast-growing states (Arizona, Georgia 2010s). The 2030 projections may miss:

- Post-pandemic migration reversal from Sun Belt states (if remote work normalization reverses the 2020–2022 domestic migration surge).
- Climate-driven migration from vulnerable coastal areas (FL, TX Gulf Coast).
- Immigration policy effects on Hispanic growth rates (major driver of TX, AZ, FL growth).

We recommend treating the Q.4 seat allocations as planning inputs with uncertainty ranges (Table 2) rather than point predictions. Infrastructure preparation should proceed for the median projection while maintaining flexibility for adjacent seat allocations in the three boundary states.

The author thanks the Census Bureau's Population Division for public access to the 2023 National Population Projections and the Defense Manpower Data Center for 2022 state-of-legal-residence counts used for the overseas adjustment. All errors are the author's own.

References

U.S. Census Bureau. 2023 national population projections: Main series, 2023. Medium-series (50th percentile) projections by state through 2060. Available from `https://www.census.gov/newsroom/press-kits/2023/national-population-projections.html`.